EXPERIMENT 2

CHARACTERISTICS STUDY OF SIDE BAND POWER AND FREQUENCIES IN AM USING MATLAB

```
clc;
clear;
close all;
%% ============================================================
% CHARACTERISTICS STUDY OF SIDEBAND POWER AND FREQUENCIES IN AM
% USING MATLAB
% ============================================================
%% 1. INITIAL PARAMETERS
Fs = 1e6;            % Sampling frequency = 1 MHz
T  = 0.01;           % Simulation time = 10 ms
t = 0:1/Fs:T-1/Fs;
% Carrier and message parameters
fc = 100e3;          % Carrier frequency = 100 kHz
fm = 10e3;           % Message frequency = 10 kHz
Ac = 1;              % Carrier amplitude
Am = 1;              % Message amplitude
R = 50;              % Load resistance = 50 ohms
%% 2. MODULATION INDEX
```

```
mu = Am/Ac;

fprintf('\n');

fprintf('=================================================\n');

fprintf('       AM SIDEBAND POWER AND FREQUENCY STUDY\n');

fprintf('=================================================\n');

fprintf('\nBasic Parameters:\n');

fprintf('Carrier Frequency       = %.2f kHz\n',fc/1000);

fprintf('Message Frequency       = %.2f kHz\n',fm/1000);

fprintf('Carrier Amplitude       = %.2f V\n',Ac);

fprintf('Message Amplitude       = %.2f V\n',Am);

fprintf('Load Resistance         = %.2f Ohm\n',R);

fprintf('Modulation Index        = %.2f\n',mu);


%% 3. MESSAGE SIGNAL

message = Am*cos(2*pi*fm*t);

%% 4. CARRIER SIGNAL

carrier = Ac*cos(2*pi*fc*t);

%% 5. AM SIGNAL

AM_signal = Ac*(1 + mu*cos(2*pi*fm*t)) ...

          .* cos(2*pi*fc*t);

%% =========================================================

% 6. TIME DOMAIN PLOTS

% =========================================================
```

```
% Message signal
figure;
plot(t*1000,message,'LineWidth',1.2);
xlabel('Time (ms)');
ylabel('Amplitude (V)');
title('Message Signal');
grid on;
xlim([0 1]);
% Carrier signal
figure;
plot(t*1000,carrier,'LineWidth',1);
xlabel('Time (ms)');
ylabel('Amplitude (V)');
title('Carrier Signal');
grid on;
xlim([0 0.2]);
% AM signal
figure;
plot(t*1000,AM_signal,'LineWidth',1);
xlabel('Time (ms)');
ylabel('Amplitude (V)');
title('Amplitude Modulated (AM) Signal');
grid on;
```

```
xlim([0 1]);
%% ============================================================
% 7. AM SPECTRUM
% ============================================================
N = length(AM_signal);
AM_fft = fft(AM_signal);
AM_fft_shifted = fftshift(AM_fft);
f = (-N/2:N/2-1)*(Fs/N);
AM_magnitude = abs(AM_fft_shifted)/N;
figure;
plot(f/1000,AM_magnitude,'LineWidth',1.2);
xlabel('Frequency (kHz)');
ylabel('Magnitude');
title('AM Frequency Spectrum');
grid on;
xlim([70 130]);
%% ============================================================
% 8. FIND USB AND LSB FREQUENCIES
% ============================================================
f_USB = fc + fm;
f_LSB = fc - fm;
fprintf('\n');
fprintf(' ideband Frequencies:\n');
```

```
fprintf('LSB Frequency          = %.2f kHz\n',f_LSB/1000);

fprintf('Carrier Frequency      = %.2f kHz\n',fc/1000);

fprintf('USB Frequency          = %.2f kHz\n',f_USB/1000);

%% ============================================================

% 9. BANDWIDTH

% ============================================================

BW_AM = 2*fm;

fprintf('\n');

fprintf('AM Bandwidth:\n');

fprintf('BW = 2fm = %.2f kHz\n',BW_AM/1000);

%% ============================================================

% 10. CARRIER POWER

% ============================================================

Pc = Ac^2/(2*R);

fprintf('\n');

fprintf('Carrier Power:\n');

fprintf('Pc = %.6f W\n',Pc);

%% ============================================================

% 11. SIDEband POWER

% ============================================================

P_USB = (mu^2 * Pc)/4;

P_LSB = (mu^2 * Pc)/4;

P_SB_total = P_USB + P_LSB;
```

```
%% ============================================================

% 12. TOTAL AM POWER

% ============================================================

P_total = Pc + P_SB_total;

%% ============================================================

% 13. POWER EFFICIENCY

% ============================================================

efficiency = (P_SB_total/P_total)*100;

%% ============================================================

% 14. DISPLAY POWER RESULTS

% ============================================================

fprintf('\n');

fprintf('Power Calculations:\n');

fprintf('Carrier Power           = %.6f W\n',Pc);

fprintf('USB Power               = %.6f W\n',P_USB);

fprintf('LSB Power               = %.6f W\n',P_LSB);

fprintf('Total Sideband Power    = %.6f W\n',P_SB_total);

fprintf('Total AM Power          = %.6f W\n',P_total);

fprintf('Transmission Efficiency  = %.2f %%\n',efficiency);

%% ============================================================

% 15. MODULATION INDEX STUDY

% ============================================================

mu_values = 0.1:0.1:1.0;
```

```
P_USB_values = zeros(size(mu_values));
P_LSB_values = zeros(size(mu_values));
P_SB_values = zeros(size(mu_values));
P_total_values = zeros(size(mu_values));
Efficiency_values = zeros(size(mu_values));
for k = 1:length(mu_values)
    m_index = mu_values(k);
    P_USB_values(k) = ...
        (m_index^2 * Pc)/4;
    P_LSB_values(k) = ...
        (m_index^2 * Pc)/4;
    P_SB_values(k) = ...
        P_USB_values(k) + P_LSB_values(k);
    P_total_values(k) = ...
        Pc + P_SB_values(k);
    Efficiency_values(k) = ...
        (P_SB_values(k)/P_total_values(k))*100;
end
%% ========================================================
% 16. DISPLAY MODULATION INDEX TABLE
% ========================================================
fprintf('\n');
fprintf('=================================================\n');
```

```
fprintf('      POWER VS MODULATION INDEX\n');
fprintf('=================================================\n');
fprintf('\n');
fprintf(' Mu     USB Power    LSB Power    Total SB     Total AM\n');
fprintf('-------------------------------------------------\n');
for k = 1:length(mu_values)
    fprintf(' %.1f     %.6f     %.6f     %.6f     %.6f\n', ...
        mu_values(k), ...
        P_USB_values(k), ...
        P_LSB_values(k), ...
        P_SB_values(k), ...
        P_total_values(k));

end
fprintf('-------------------------------------------------\n');
%% ============================================================
% 17. USB POWER VS MODULATION INDEX
% ============================================================
figure;
plot(mu_values,P_USB_values,'o-','LineWidth',1.5);
xlabel('Modulation Index (\mu)');
ylabel('USB Power (W)');
title('USB Power vs Modulation Index');
```

```
grid on;
%% ============================================================
% 18. LSB POWER VS MODULATION INDEX
% ============================================================

figure;
plot(mu_values,P_LSB_values,'s-','LineWidth',1.5);
xlabel('Modulation Index (\mu)');
ylabel('LSB Power (W)');
title('LSB Power vs Modulation Index');
grid on;
%% ============================================================
% 19. TOTAL SIDEBAND POWER VS MODULATION INDEX
% ============================================================
figure;
plot(mu_values,P_SB_values,'^-','LineWidth',1.5);
xlabel('Modulation Index (\mu)');
ylabel('Total Sideband Power (W)');
title('Total Sideband Power vs Modulation Index');
grid on;
%% ============================================================
% 20. TOTAL AM POWER VS MODULATION INDEX
% ============================================================
```

```
figure;
plot(mu_values,P_total_values,'o-','LineWidth',1.5);
xlabel('Modulation Index (\mu)');
ylabel('Total AM Power (W)');
title('Total AM Power vs Modulation Index');
grid on;
%% ============================================================
% 21. TRANSMISSION EFFICIENCY
% ============================================================
figure;
plot(mu_values,Efficiency_values,'d-','LineWidth',1.5);
xlabel('Modulation Index (\mu)');
ylabel('Efficiency (%)');
title('AM Transmission Efficiency vs Modulation Index');
grid on;
%% ============================================================
% 22. COMPARE USB AND LSB POWER
% ============================================================
figure;
plot(mu_values,P_USB_values,'o-','LineWidth',1.5);
hold on;
plot(mu_values,P_LSB_values,'s-','LineWidth',1.5);
xlabel('Modulation Index (\mu)');
```

```
ylabel('Power (W)');
title('USB and LSB Power Comparison');
legend('USB Power','LSB Power');
grid on;
%% ============================================================
% 23. AM SPECTRUM FOR DIFFERENT MODULATION INDICES
% ============================================================
mu_plot = [0.25 0.5 0.75 1.0];
figure;
for k = 1:length(mu_plot)
    mu_temp = mu_plot(k);
    AM_temp = Ac*(1 + ...
        mu_temp*cos(2*pi*fm*t)) ...
        .* cos(2*pi*fc*t);
    spectrum_temp = fftshift(fft(AM_temp));
    magnitude_temp = ...
        abs(spectrum_temp)/N;
    subplot(4,1,k);
    plot(f/1000,magnitude_temp);
    xlabel('Frequency (kHz)');
    ylabel('Magnitude');
    title(['AM Spectrum, \mu = ', ...
        num2str(mu_temp)]);
```

```
    grid on;
    xlim([70 130]);
end
%% ============================================================
% 24. ENVELOPE OF AM SIGNAL
% ============================================================
envelope_upper = Ac*(1 + mu*cos(2*pi*fm*t));
envelope_lower = -Ac*(1 + mu*cos(2*pi*fm*t));
figure;
plot(t*1000,AM_signal);
hold on;
plot(t*1000,envelope_upper,'--','LineWidth',1.2);
plot(t*1000,envelope_lower,'--','LineWidth',1.2);
xlabel('Time (ms)');
ylabel('Amplitude (V)');
title('AM Signal and Its Envelope');
legend('AM Signal','Upper Envelope','Lower Envelope');
grid on;
xlim([0 1]);
%% ============================================================
% 25. IMPORTANT RESULTS AT 100% MODULATION
% ============================================================
mu_100 = 1;
```

```
P_USB_100 = (mu_100^2 * Pc)/4;

P_LSB_100 = (mu_100^2 * Pc)/4;

P_SB_100 = P_USB_100 + P_LSB_100;

P_total_100 = Pc + P_SB_100;

efficiency_100 = ...

   (P_SB_100/P_total_100)*100;

fprintf('\n');

fprintf('=====================================================\n');

fprintf('         RESULTS FOR 100%% MODULATION\n');

fprintf('=====================================================\n');

fprintf('Carrier Power           = %.6f W\n',Pc);

fprintf('USB Power               = %.6f W\n',P_USB_100);

fprintf('LSB Power               = %.6f W\n',P_LSB_100);

fprintf('Total Sideband Power    = %.6f W\n',P_SB_100);

fprintf('Total AM Power          = %.6f W\n',P_total_100);

fprintf('Transmission Efficiency  = %.2f %%\n',efficiency_100);

fprintf('=====================================================\n');

%% ========================================================

% 26. FINAL CONCLUSION

% ========================================================

fprintf('\n');

fprintf('CONCLUSION:\n');

fprintf(['1. USB frequency = fc + fm.\n']);
```

```
fprintf(['2. LSB frequency = fc - fm.\n']);

fprintf(['3. AM bandwidth = 2fm.\n']);

fprintf(['4. Sideband power increases with mu^2.\n']);

fprintf(['5. At 100%% modulation, total sideband power = Pc/2.\n']);

fprintf(['6. At 100%% modulation, total AM power = 1.5Pc.\n']);

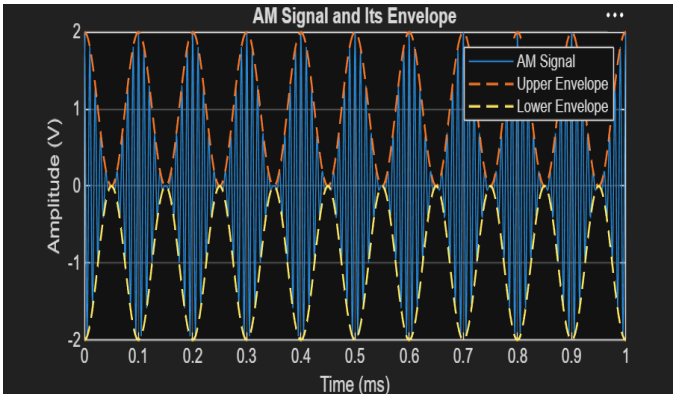
fprintf(['7. Maximum theoretical AM efficiency = 33.33%%.\n']);

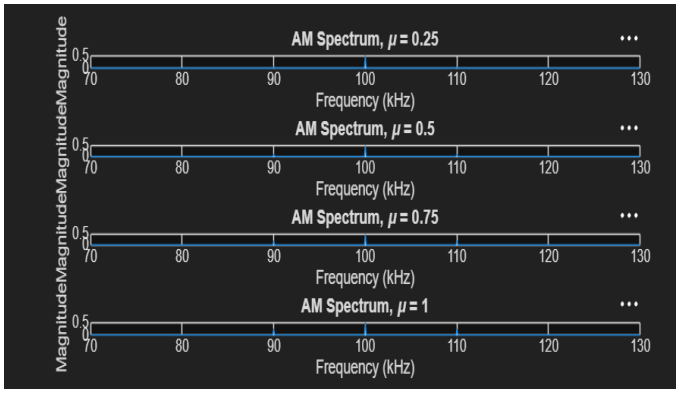
fprintf('\n==================================================\n');

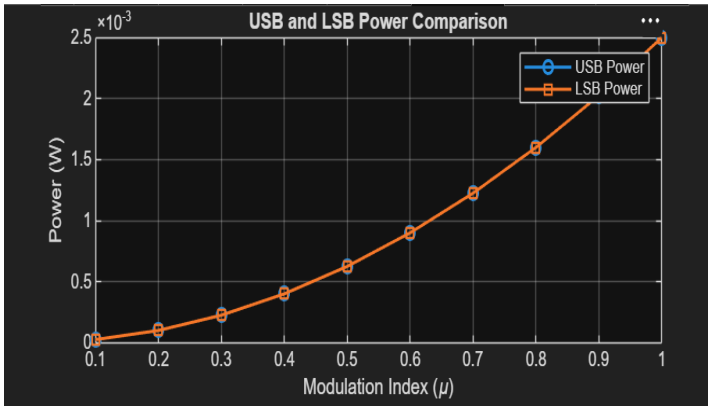
fprintf('            SIMULATION COMPLETED\n');

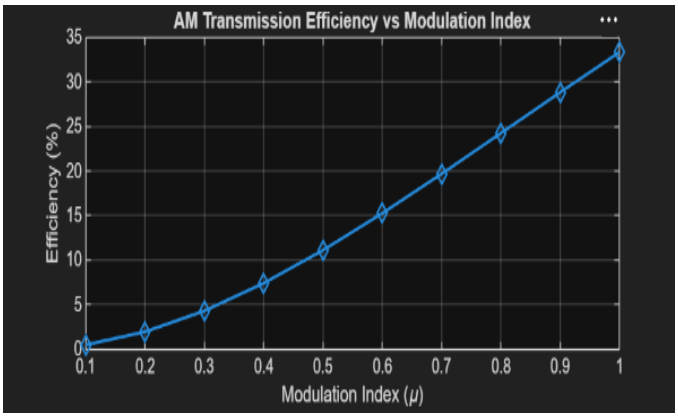
fprintf('==================================================\n'):
```

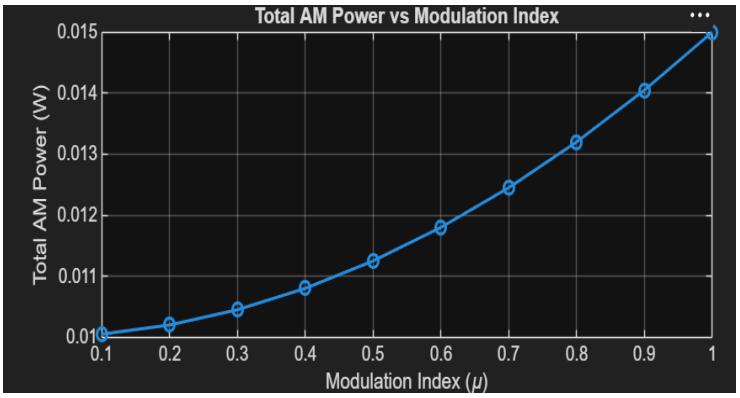
OUTPUT

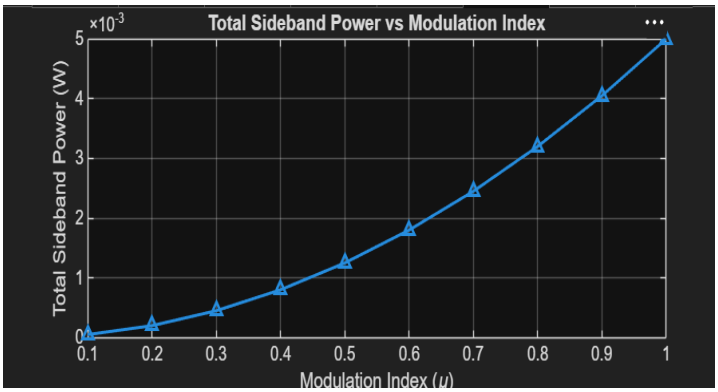


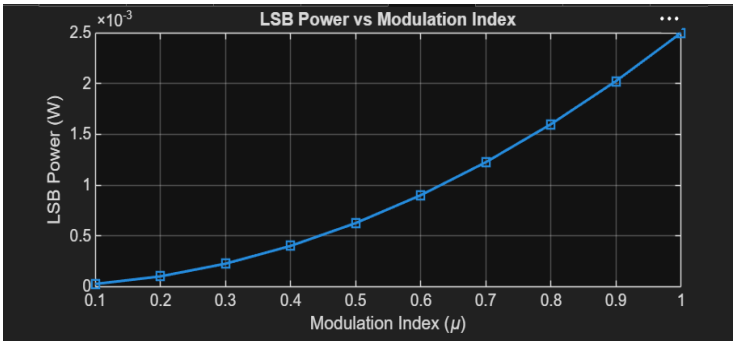


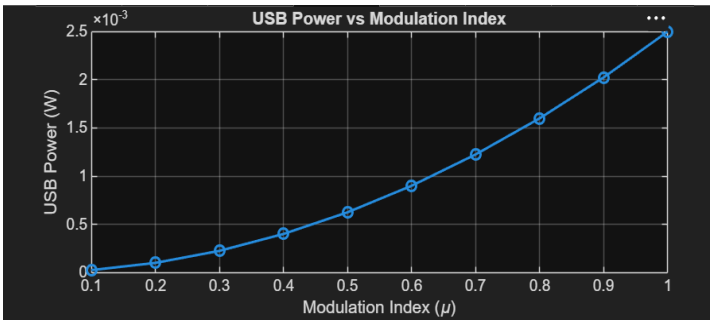


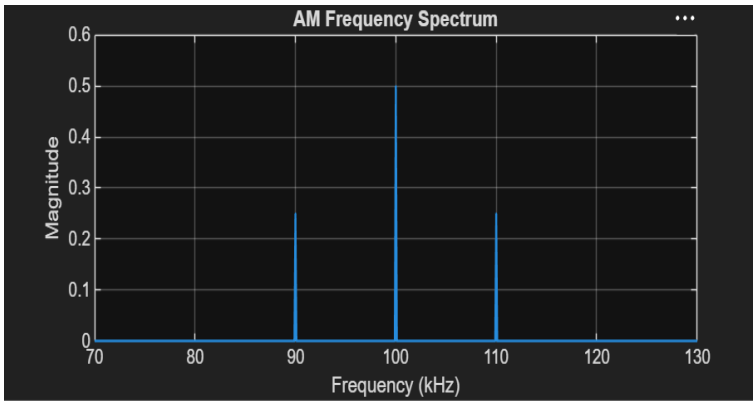


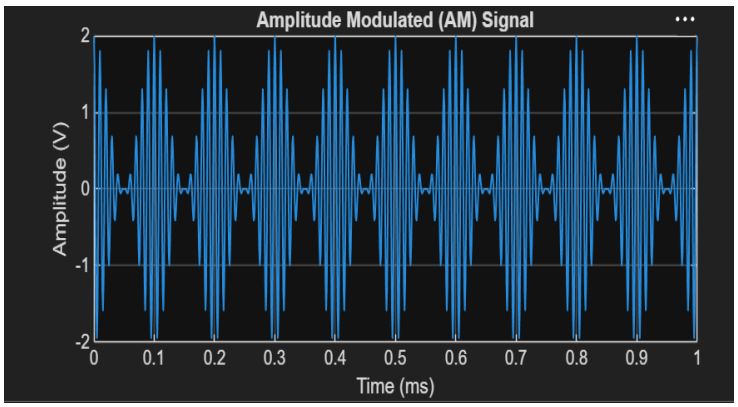

Total AM Power vs Modulation Index
Total AM Power (W)
Modulation Index (μ)

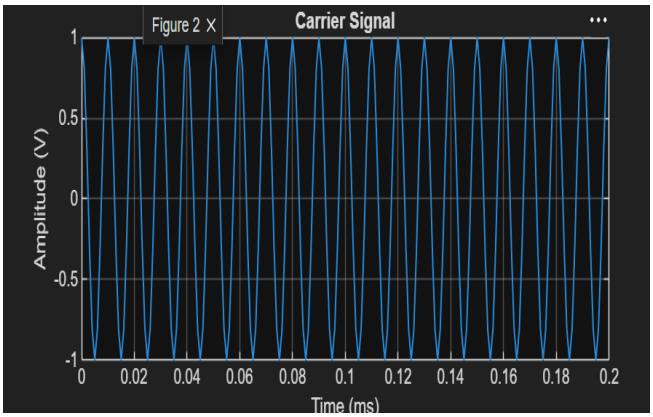

Total Sideband Power vs Modulation Index
×10-3
Total Sideband Power (W)
Modulation Index (μ)

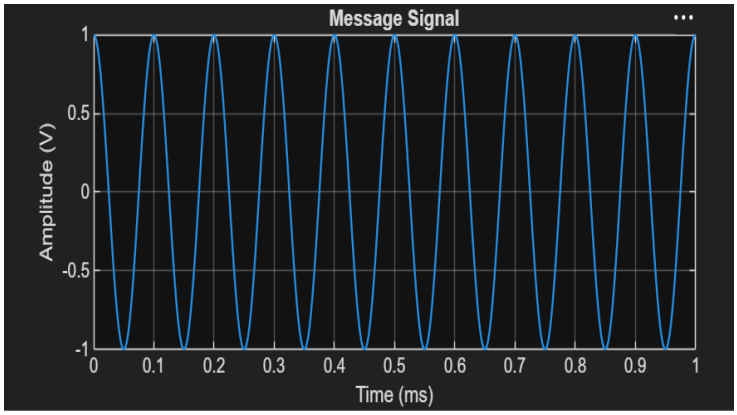

LSB Power vs Modulation Index
×10-3
LSB Power (W)
Modulation Index (μ)


USB Power vs Modulation Index
×10-3
USB Power (W)
Modulation Index (μ)


AM Frequency Spectrum
Magnitude
Frequency (kHz)


Amplitude Modulated (AM) Signal
Amplitude (V)
Time (ms)


Figure 2
Carrier Signal
Amplitude (V)
1
0.5
0
-0.5
-1
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
Time (ms)


Message Signal
Amplitude (V)
1
0.5
0
-0.5
-1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Time (ms)